**Diagnostic Report generated by gemini-2.5-pro**

# Diagnostic Analysis of Run_06

## 1. Summary of Key Deviations

The analysis of Run_06 shows a very specific and isolated set of anomalies.

- **Flow_PRV (Critical):** The signal is in a state of **critical saturation** (+1095% to +11246%) across all operational states. This is a persistent and non-physical reading.
- **Flow_Pump (Minor):** A very slight positive **offset** (approx. +4%) is observed consistently from state s2 onwards.
- **Flow_Bypass (Minor):** A minor positive **offset** (approx. +3%) appears only in the high-throttle and later states (s5-s6).
- **All Other Signals:** All other signals, including all pressures (`p_Pump`, `p_FMU`, `p_Combustion`, `p_Shut`) and all engine flows (`Flow_Engine1-4`), show **no significant deviation** from healthy operation throughout the entire test run.

## 2. Root-Cause Hypotheses

| Rank | Suspected Component | Fault Type | Supporting Evidence |
|---|---|---|---|
| 1 | **Flow_PRV Sensor** | **Sensor Saturation / Total Failure** | The deviation is extreme (+11000%) and physically impossible. Crucially, there are **no corresponding physical effects**—system pressures and engine flows remain perfectly healthy. This is the classic signature of a complete, isolated sensor failure. |
| 2 | **Flow_Pump Sensor** | **Minor Calibration Error** | The slight positive offset on **Flow_Pump** (+4%) is minor and not reflected in any pressure increases or significant flow changes downstream. This could indicate a minor calibration error in just this one sensor. |

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|---|---|---|---|
| 3 | **Pressure Relief Valve (PRV)** | **Mechanical Jam (Closed)** | A jammed-shut PRV would not open under any condition. However, if the system never reaches the PRV's cracking pressure, this fault would have no symptoms. The saturated sensor reading makes it impossible to know the actual flow, but the lack of pressure anomalies makes a physical valve fault unlikely. |
| 4 | **Main Fuel Pump** | **Minor Over-delivery** | The pump might be delivering slightly more flow (+4%), causing the minor offset on the **Flow_Pump** sensor. However, without a corresponding rise in system pressure, this is less likely to be a physical issue and more likely to be a sensor measurement anomaly. |
| 5 | **Wiring / Electrical Fault** | **Signal Cross-Talk / Grounding Issue** | An electrical fault specific to the **Flow_PRV** sensor circuit could cause a permanently saturated signal without affecting the physical system. This is a plausible explanation for the isolated nature of the critical fault. |

## 3. Most Probable Hypothesis

The most probable root cause is a **Total Sensor Failure of the Flow_PRV Sensor.**

**Reasoning:** The evidence for this hypothesis is overwhelming. The `Flow_PRV` reading is critically saturated from the very first state to the last, which is physically implausible. Most importantly, this massive anomaly occurs in complete isolation. None of the other system signals show any physically consistent secondary effects. If the PRV were actually dumping massive amounts of flow, system pressures and engine flows would plummet, which they do not. The system is, for all intents and purposes, operating perfectly, with the sole exception of this one erroneous sensor reading. The minor offsets on `Flow_Pump` and `Flow_Bypass` are likely within nominal variance or represent very minor, unrelated calibration drifts, not a true fault condition.